SIMATS ENGINEERING

CONTINOUS INTERNAL ASSESSMENT- SHORT ANSWER TEXT

COURSE NAME: DATABASE MANAGEMENT SYSTEMS
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Questions:05 **Total marks:100**

Annexure A

SHORT ANSWER TEXT

Code of Conduct:

I Swaharishwaran 192126O142 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Swaharish

Vehicle Management System - EER Diagram.

1) Identify the supertype and subtypes.

Supertype : Vehicle

Subtype : Car and truck.

Vehicle is the superiface that contains common attributes.

2) Common attributes of vehicle :

1. Vehicle - ID → Primary key
2. Registration - Number
3. manufactur
4. model
5. Year
6. Purchase - price.

3) Specialized Attributes of car and Truck.

Car Attributes -

NO - of - Door

NO - of - fuel - type

Seating - capacity

Truck Attributes.

Load - capacity

Truck - type

NO - of - Axles.

4) Primary key and Inheritance Vehicle - ID is the primary key of vehicle supertype. Both car and Truck inherit this primary key along with the common attributes of vehicles.

5) ISA specialization:

→ car inhobit all common attributes for vehicle.

→ Truck inherit all attributes from vehicle.

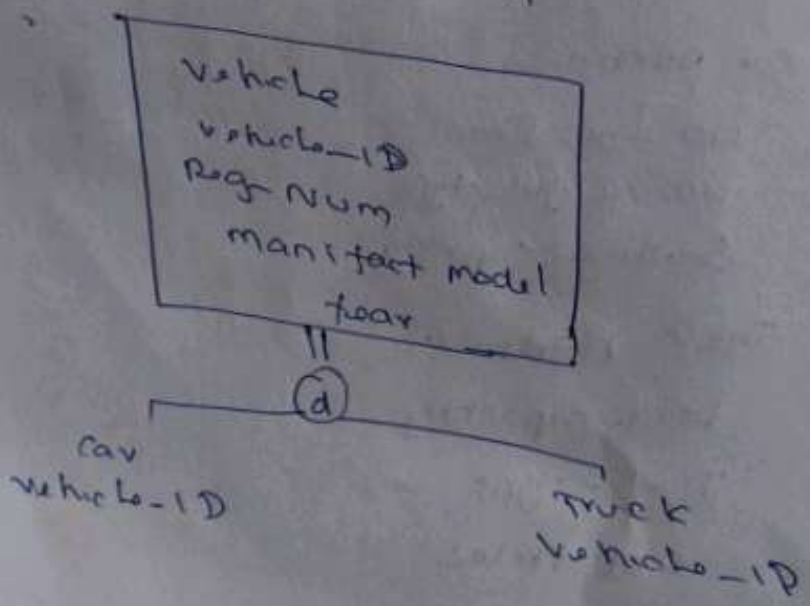
6) Disjoint or Oylapping:

* Vehicle can belong to either car or truck subject Disjoint.

7) Total or partical.

The specilisation is total

8)



Bank management System:

① The major entities are Customer, Account, Branch, Employes, Transaction Loan, Loan- Payment.

② & ③

Attributes and primary key

Customer: Customer-ID, Customer name, Address, Dress, Email.

Account: Account-NO

Branch: Branch-ID

Employs: Employes-ID

Transaction: transaction-ID

Loan_: Loan-ID

④ & ⑤

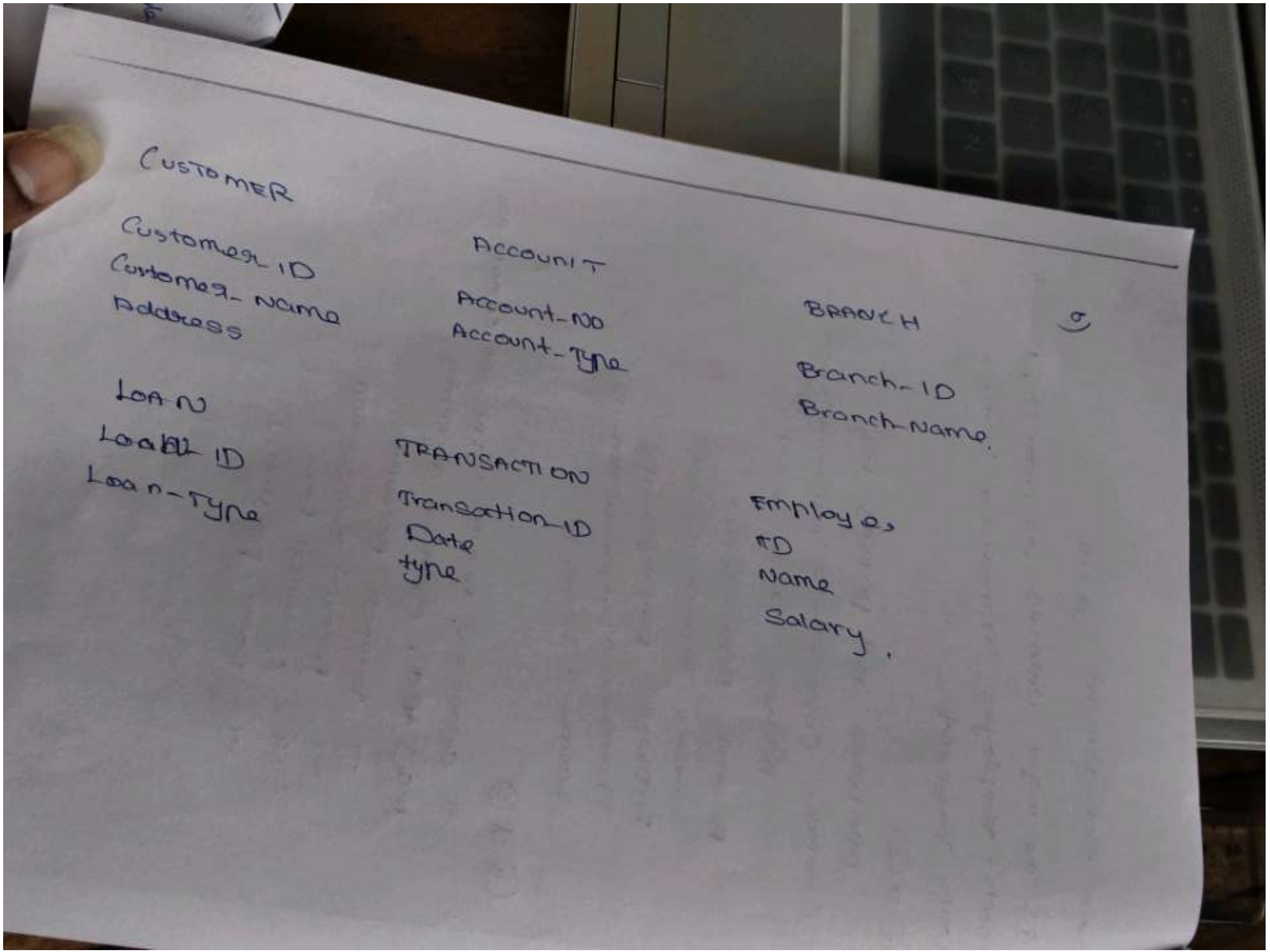
→ Customer Account: 1:N: one customer may hold. multiple accounts.

→ Branch- Account

→ Account transaction

→ Customer Loan

→ Loan - payment



CUSTOMER

Customer_ID
Customer_name
Address

ACCOUNT

Account_no
Account_type

BRANCH

Branch_ID
Branch_Name.

LOAN

Loan_ID
Loan_Type

TRANSACTION

Transaction_ID
Date
type

Employee
ID
Name
Salary.

E-commerce website:

① Major Entities

* Customer, Products, Category, order, payment.

② Attributes and Primary key.

→ Customer: Customer-ID
→ Product: Product-ID
→ Category: Category-ID
→ Delivery: Delivery-ID

③ order- product relationship.

* order and product have many to many relationship.

④ Product is Supertype.

Subtype are
* electronics
* clothing
* Grocery.

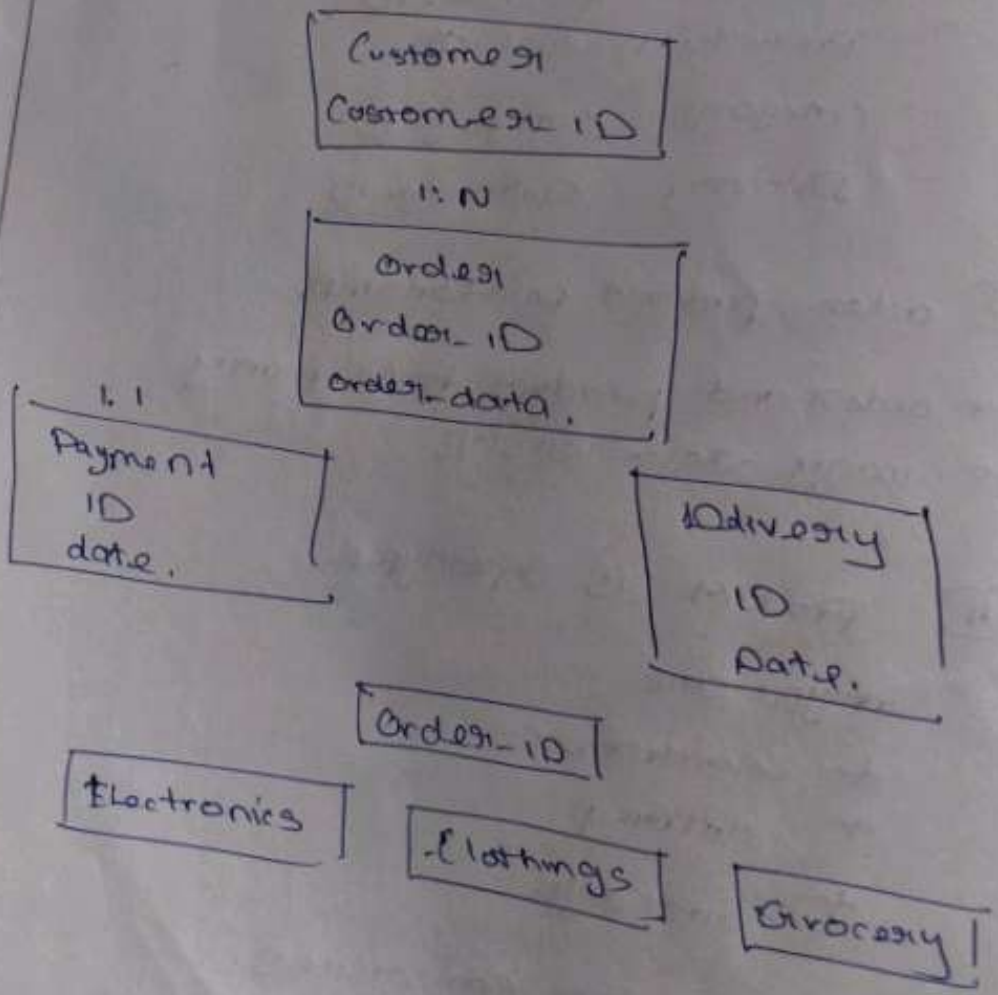
5.b Specialization constrains:

* Disjoint → only belong three Subtypes
* total → Every product belong to three Subtype.

7 8. Relationship and Cardinalities

Customer - order - 1 : N
order - Payment - 1 : 1
order - Delivery - 1 : 1

9. ER DIAGRAM.



Hospital Managment System.

1. Major entities.

* The major entities are staff, Doctor, Nurse, Admin, Department, Appointment.

2-4. Staff specialisation and attributes.

Staff	Doctor
Staff-ID	Staff-ID
Name	Specilisation
Address	License Number.
Phone	
Salary	

5-7. Week entity and identifying.

Medical record is each entity relashin.

Partial key - Record - Date

Identifing relationship - HAS - Record.

8. 9. Department — Relationship.

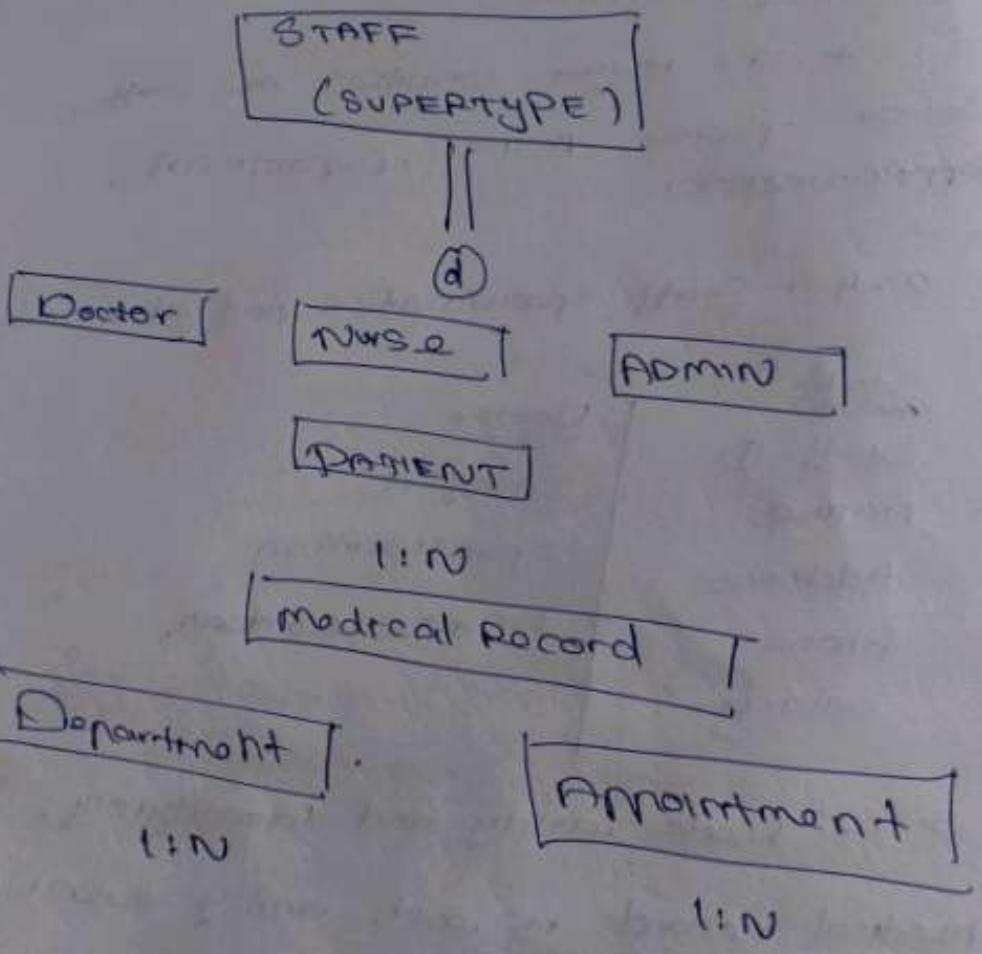
Dept - Doctor 1: N

Patient - Appoin 1: N

Doctor - Appoin 1: N.

Patient - medical - record 1: N

EER DIAGRAM :


STAFF
(SUPERTYPE)
d
Doctor
Nurse
ADMIN
PATIENT
1:N
Medical Record
Department
1:N
Appointment
1:N

Database Administrator:

① A Database administration is responsible for installing, configuring, security, maintaing and maintainy.

② The DBA helps design database schemas, tables, relationships, indexes, constraints. DBA selects suitable database.

3) The DBA protects data from access by implementing authentication, encryption and auditing.

4) The DBA schedules and verifies regular database backup, Recovery procedures are developed.

5) The DBA monitors query execution time, CPU useage, memory consumption, disk.

6) The DBA maintains data accuracy and consistency by enforcing primary key unique constraints & consistancies between related tables.

7. DBA manages database storage capacity transaction key upgrading software & maintaing & disk space.

8. DBA identifies and resolves database error, connection failure, deadlocks, corruption & performance problem.

9. User and privilege management DBA creats, modifies, disable database user accounts.

10. The DBA ensures database availability through monitory, replication & clustering and failure mechanism & where appropriate.